DBMS Homework

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Due date: 2025.11.09
Submission date: 2025.11.09

QUERY#1

```
-- HOMEWORK
-- HWQ1 For each department,dep number,  number of employees, and their avarage salary
SELECT dnumber, COUNT(*) staff_count, round(AVG(salary)) avg_salary
FROM department, employee
WHERE dno=dnumber
GROUP BY Dnumber;
```

OUTPUT#1

Query Result 1 | Query Result 2 | Query Result 3 | Q

SQL | All Rows Fetched: 6 in 0.343 seconds

	DNUMBER	STAFF_COUNT	AVG_SALARY
1	1	1	55000
2	5	4	33250
3	4	3	31000
4	6	8	60000
5	7	10	63450
6	8	14	40821

QUERY#2

```
-- HWQ2 for each project, retrieve the project number, the project name, name,
---the number of employees who WOrk on that project
SELECT pnumber, pname, COUNT(works_on.essn) staff_count
FROM project, works_on
WHERE pnumber=pno
GROUP BY pnumber, pname;
```

OUTPUT#2

Query Result

SQL | All Rows Fetched: 11 in 0.106 seconds

	PNUMBER	PNAME	STAFF_COUNT
1	1	ProductX	2
2	2	ProductY	3
3	3	ProductZ	2
4	10	Computerization	3
5	20	Reorganization	3
6	30	Newbenefits	3
7	61	OperatingSystems	9
8	62	DatabaseSystems	8
9	63	Middleware	4
10	91	InkjetPrinters	8
11	92	LaserPrinters	3

QUERY#3

```
-- HWQ3 for each project on which more than two employees WOrk, retrieve
-- the project number and name, and the number of employees who work on project
SELECT pnumber, pname, COUNT(works_on.essn) staff_count
FROM project, works_on
WHERE pnumber=pno
GROUP BY pnumber, pname
HAVING staff_count > 2;
```

OUTPUT#3

Query Result 2

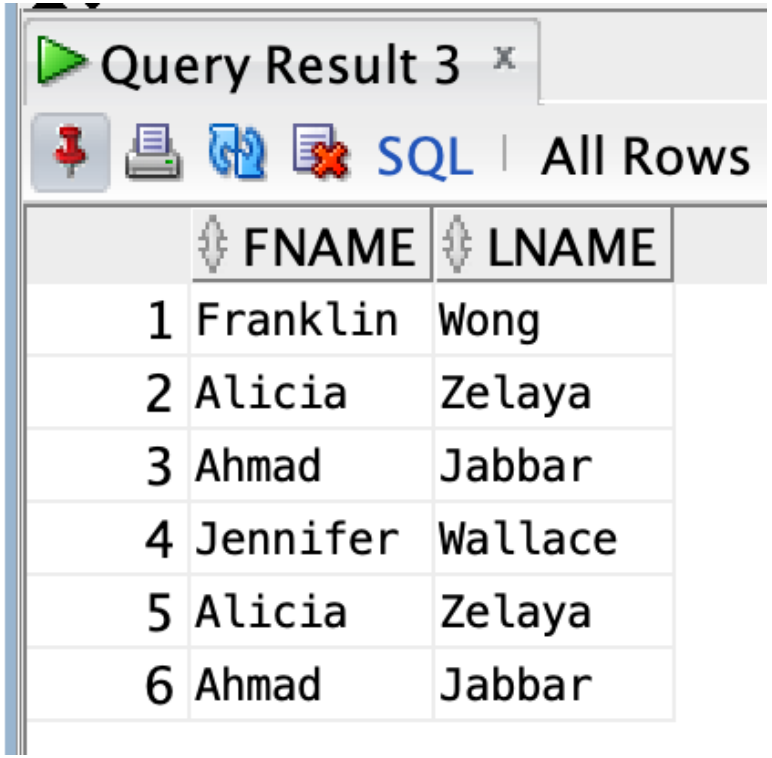
SQL | All Rows Fetched: 9 in 0.125 seconds

	PNUMBER	PNAME	STAFF_COUNT
1	2	ProductY	3
2	10	Computerization	3
3	20	Reorganization	3
4	30	Newbenefits	3
5	61	OperatingSystems	9
6	62	DatabaseSystems	8
7	63	Middleware	4
8	91	InkjetPrinters	8
9	92	LaserPrinters	3

QUERY#4

```
-- HWQ4 retrieve the name of each employee who works on the projects controlled
-- by department number 4
SELECT Fname, Lname
FROM employee E, works_on W, project P
WHERE E.ssn=W.essn AND W.pno=P.pnumber AND P.dnum=4;
```

OUTPUT#4



	FNAME	LNAME
1	Franklin	Wong
2	Alicia	Zelaya
3	Ahmad	Jabbar
4	Jennifer	Wallace
5	Alicia	Zelaya
6	Ahmad	Jabbar

QUERY#5

```
-- HwQ5 retrieve the name of each employee who works on ALf the projects
-- controlled by department number 4

SELECT concat(e1.Fname, ' ', e1.Lname) fullname
FROM employee e1
WHERE NOT EXISTS (
(SELECT Pnumber FROM project WHERE dnum=4)
MINUS
(SELECT Pnumber FROM works_on, project WHERE essn=e1.ssn AND dnum=4)
);
```

OUTPUT#5

Query Result 4

SQL | All Rows Fetched: 38 in 0.112 seconds

	FULLNAME
26	Nandita Ball
27	Jill Jarvis
28	Lyle Leslie
29	Billie King
30	Jon Kramer
31	Ray King
32	Gerald Small
33	Arnold Head
34	Helga Pataki
35	Naveen Drew
36	Carl Reedy
37	Sammy Hall
38	Red Bacher